

Keno: generative noise subtraction for template-free gravitational-wave burst search

Panagiotis Minoglou*

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Abstract

Keno is a research software prototype for searching short gravitational-wave bursts without assuming a known waveform shape. A neural network learns to predict detector noise from whitened sensor data; that prediction is subtracted; and the leftover residual is searched with an energy-based statistic, a two-detector timing check, and a simple consistency veto.

On synthetic bursts injected into real LIGO noise, at a 1% false-alarm rate (about one false trigger per hundred noise-only tests), Keno residual search reaches near 100% detection efficiency from signal-to-noise ratio (SNR) 2 to 12. Searching the same data without subtraction, or with a deliberately wrong matched-filter template, is less efficient. On published event times from public catalogs, a coherent two-detector residual search recovers 9 of 18 dual-detector events after rejecting a known glitch near GW170817.

Keno is meant to complement morphology-specific binary black hole classifiers, not replace them. The contribution is a reproducible software path for unmodeled residual search.

1 Introduction

LIGO and Virgo record a time series called *strain*: a dimensionless measure of how much spacetime stretches as gravitational waves pass the detector. In software terms, strain is a 1D signal sampled at high rate (here 4096 Hz). Before analysis it is usually *whitened*: filtered so that stationary background noise has roughly unit variance across frequency, which makes short anomalies easier to see.

Search methods fall into two complementary families:

- **Morphology-specific / template-based.** If the expected signal shape is known (for example a binary black hole merger chirp), one can match-filter against templates or train a classifier on that family. AresGW [Nousi et al., 2023, Koloniari et al., 2025] is a successful example of the classifier approach for black hole mergers.
- **Template-free / unmodeled.** If the shape is unknown (short rings, noise-like bursts, or other transients), one instead looks for localized excess energy that is consistent across detectors, without assuming a particular waveform.

Keno belongs to the second family. It adds a generative denoising step before the energy search: a neural network predicts instrumental noise $\hat{N}$ and forms a cleaned residual

$$S_{\text{clean}} = S_{\text{raw}} - \hat{N}. \quad (1)$$

The residual is then searched with excess power (an STFT peak-energy statistic), an H1+L1 lag scan (Hanford and Livingston detectors), and an envelope-consistency veto that rejects loud single-detector artifacts (*glitches*).

Keno was developed as an independent research prototype by a software engineer. It is not a LIGO collaboration product and is not presented as a production search pipeline.

*Independent researcher / software engineer. Corresponding author. E-mail: panosmng97@gmail.com

Keno production detection path

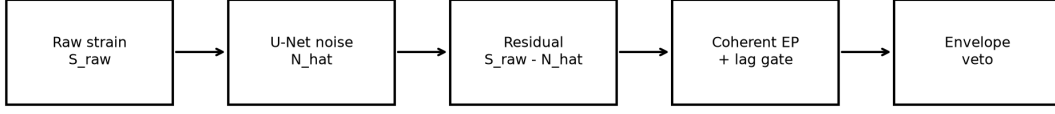


Figure 1: Keno detection path: predict noise, subtract, search residual excess power, then check H1+L1 timing consistency with an envelope veto.

Relation to morphology-specific classifiers. Systems such as AresGW [Nousi et al., 2023] answer: “does this segment look like a black hole merger I was trained on?” Keno answers: “after I remove the noise I think I understand, does excess energy remain?” Those are different jobs. Keno is intended as a complementary contribution for unmodeled bursts.

Claim. In this prototype, generative subtraction followed by residual excess-power search improves recovery of unknown-morphology bursts at controlled false-alarm rate, relative to (i) excess power on the unsubtracted strain and (ii) matched filtering with a deliberately wrong template.

2 Background terms (for software readers)

SNR Signal-to-noise ratio. Roughly how loud the injected burst is relative to the background; higher SNR is easier to detect.

FAR False-alarm rate. The fraction of pure-noise trials that still cross the detection threshold. A 1% FAR means about 1 in 100 noise-only windows is counted as a false trigger.

Matched filter Correlate the data against a known waveform template. Very powerful when the template matches the true signal; weak when it does not.

Excess power A template-free energy statistic: look for a loud time–frequency patch relative to a robust noise floor (here, the median STFT power).

Residual What remains after subtracting predicted noise from the input. Ideally background shrinks and a real burst survives.

Glitch A non-astrophysical detector artifact that can look like a burst in one interferometer.

H1 / L1 LIGO Hanford and LIGO Livingston. Requiring both reduces single-detector false alarms.

3 Method

Figure 1 summarizes the detection path. Strain segments are whitened and sampled at $f_s = 4096$ Hz. Unless noted, analysis windows are $T = 4$ s ($N = 16\,384$ samples).

3.1 Generative noise model

Keno uses a 1D U-Net [Ronneberger et al., 2015] with four encoder/decoder stages (base width 16 channels, kernel size 9, GELU activations, batch normalization). In software terms this is a convolutional autoencoder-style network for 1D sequences: it maps whitened strain S_{raw} to a predicted noise waveform $\hat{N}$. The residual is

$$R = S_{\text{raw}} - \hat{N}. \quad (2)$$

Training uses cached real public LIGO background windows (GWOSC) for H1/L1. A residual loss term encourages injected short bursts to survive subtraction instead of being absorbed into $\hat{N}$. A hold-out acceptance check requires mean normalized recovery error $\text{RMS}(R - s)/\text{RMS}(s) < 0.15$ on injections with known ground-truth signal s . The checkpoint used here is frozen as 2026-07-paper-v1 (SHA256 prefix 55ce7637...).

3.2 Template-free excess-power statistic

For a residual or raw series $x(t)$, Keno computes a short-time Fourier transform (Hann window, $n_{\text{perseg}} = 256$, 75% overlap) — the usual STFT used in audio/DSP work. It forms bin power $P(f, t)$, estimates a robust noise floor as $\text{median}(P)$, and defines

$$\rho_{\text{EP}}(x) = \max_{f,t} \frac{P(f, t)}{\text{median}(P)}. \quad (3)$$

Intuitively: “how many times louder is the loudest spectrogram cell than the typical background cell?” No template correlation is required.

3.3 Single-detector residual gate

A residual threshold θ_{IFO} is calibrated on noise-only windows at a target FAR. Because imperfect subtraction can create rare large artifacts, the upper 1% of noise-only residual scores is trimmed before taking the $(1 - \text{FAR})$ percentile. In the freeze, $\theta_{\text{IFO}} \approx 4004.6$ at 1% FAR. This gate is useful for single-detector diagnostics. Dual-detector production detection uses the coherent threshold below.

3.4 Coherent dual-detector lag scan

For residuals from Hanford and Livingston, R_{H} and R_{L} , Keno scans small time shifts $|\ell|/f_s \leq 10$ ms (about the light-travel time between sites) and relative sign flips $p \in \{+1, -1\}$:

$$C_{\ell,p} = \frac{R_{\text{H}} + p \mathcal{S}_{\ell}(R_{\text{L}})}{\sqrt{2}}, \quad (4)$$

where $\mathcal{S}_{\ell}$ is a zero-padded shift. The coherent statistic is

$$\rho_{\text{coh}} = \max_{\ell,p} \rho_{\text{EP}}(C_{\ell,p}), \quad (5)$$

with best lag $\tau^* = \ell^*/f_s$. This is a software cross-check: a real astrophysical signal should line up in both detectors within a few milliseconds; a random glitch usually will not.

3.5 Envelope-consistency veto

Independent residual envelope peak times t_{H} and t_{L} are estimated from a short smoothed energy envelope (8 ms boxcar), ignoring the first/last 64 ms so edge artifacts do not dominate. Define

$$\Delta t_{\text{env}} = (t_{\text{L}} - t_{\text{H}}) \times 10^3 \quad (\text{ms}). \quad (6)$$

If $|\Delta t_{\text{env}}| > 50$ ms while the coherent score is loud, the candidate is treated as likely single-detector glitch contamination (for example the known Livingston glitch near GW170817).

3.6 Production detection rule

A dual-detector production trigger is declared if and only if

$$\rho_{\text{coh}} \geq \theta_{\text{coh}}, \quad (7)$$

$$|\tau^*| \leq 10 \text{ ms}, \quad (8)$$

$$|\Delta t_{\text{env}}| \leq 50 \text{ ms}. \quad (9)$$

The coherent threshold θ_{coh} is calibrated on dual-detector noise trials that already pass the envelope gate. With 1500 noise trials (23 envelope-gated), the 99th percentile of gated coherent EP yields $\theta_{\text{coh}} \approx 173.1$ at empirical joint FAR 0.07% (1/1500). After gating, coherent noise is much quieter than the single-detector residual threshold.

3.7 Baselines

All baselines use the same whitened windows and the same target FAR on noise-only trials:

- **Oracle matched filter** — correlate against the true injected waveform (injection studies only; an upper bound).
- **Mismatched matched filter** — correlate against a fixed sine-Gaussian template that does not match the injected family (control for wrong-template search).
- **Raw excess power** — $\rho_{\text{EP}}(S_{\text{raw}})$ with no generative subtraction.
- **BBH-trained ResNet (control)** — a compact 1D ResNet trained only on black-hole-merger-like chirps versus noise in the same cache, scored as $P(\text{signal})$ on the central 1 s. This is a local morphology-specific control in the spirit of BBH classifiers such as AresGW [Nousi et al., 2023]; it is not a published external weight file.
- **Keno residual EP** — $\rho_{\text{EP}}(R)$.
- **Keno overlap** — cosine similarity between residual and injected signal (evaluation only; needs ground truth).

4 Experiments

Injection campaigns mix short unknown morphologies (sine-Gaussian, ringdown, white-noise burst) into cached real LIGO background at equal FAR. Additional studies include a glitch stress test on public O3 Gravity Spy labels, dual-detector noise coincidence FAR, and a follow-up on published GWTC/cWB GPS times (public catalog timestamps). The catalog follow-up is a consistency check, not a discovery claim.

5 Results

5.1 Unknown-morphology efficiency

Figure 2 shows Keno residual EP at 100% efficiency across SNR 2–12 at 1% FAR. Raw excess power on unsubtracted strain is less efficient on the same injections. Mismatched matched filtering remains near $\sim 20\%$ and never reaches 50% efficiency. The BBH-trained ResNet control, evaluated on these non-BBH bursts, plateaus below saturation — expected for a morphology-specific model outside its training objective.

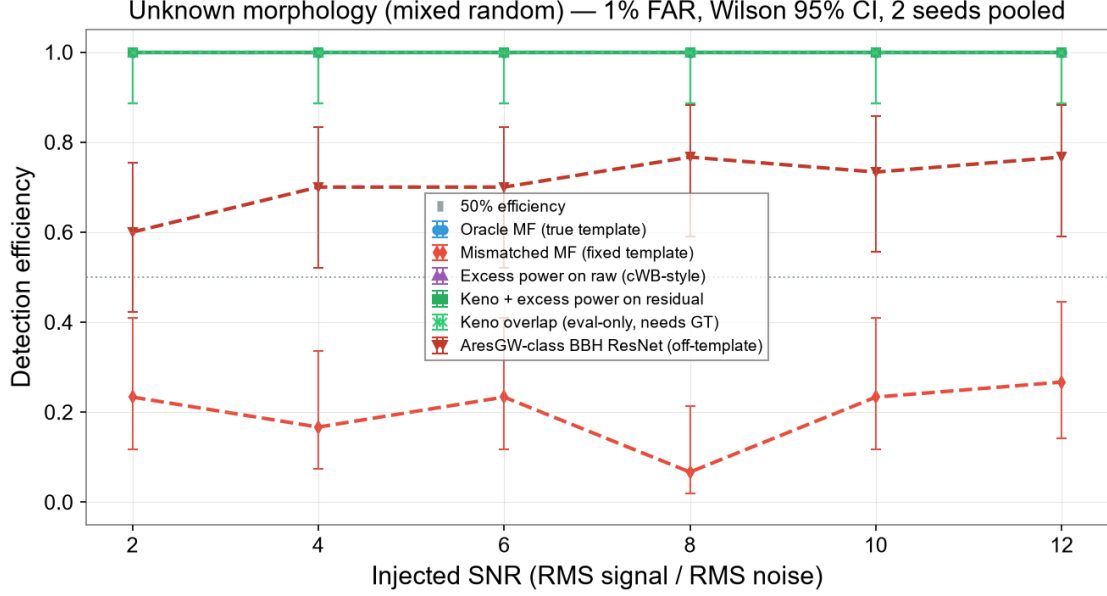


Figure 2: Detection efficiency vs injected SNR at 1% FAR on unknown morphology. Baselines include raw excess power, mismatched matched filtering, and a BBH-trained ResNet control evaluated off-template.

5.2 Coherent calibration

Envelope gating removes most dual-detector noise outliers. With 1500 noise trials (23 envelope-gated), the coherent EP threshold is 173.1 at empirical joint FAR 0.07%, separate from the single-detector residual threshold ≈ 4005 .

5.3 Catalog GPS follow-up

Production coincidence recovers 9/18 dual-detector catalog events (Figure 3). GW170817 is rejected by the envelope veto despite huge coherent EP (Livingston glitch). Contaminated recoveries are not counted as detections. A small set of cWB-only candidates remains 0/3.

6 Discussion and limitations

Keno is proposed as a complementary software path for unmodeled residual search. Morphology-specific BBH classifiers and matched-filter compact-binary pipelines [Nousi et al., 2023, Koleniari et al., 2025] remain the right tools when the target family is known.

Known limitations of this prototype include:

- weak single-detector glitch rejection; coincidence and the envelope veto do most of that work;
- residual over-subtraction risk if training is insufficient, mitigated by the recovery-error acceptance check;
- incomplete recovery of some catalog candidates in this freeze;
- development by a single engineer with a research codebase, without collaboration-scale detector characterization or production operations.

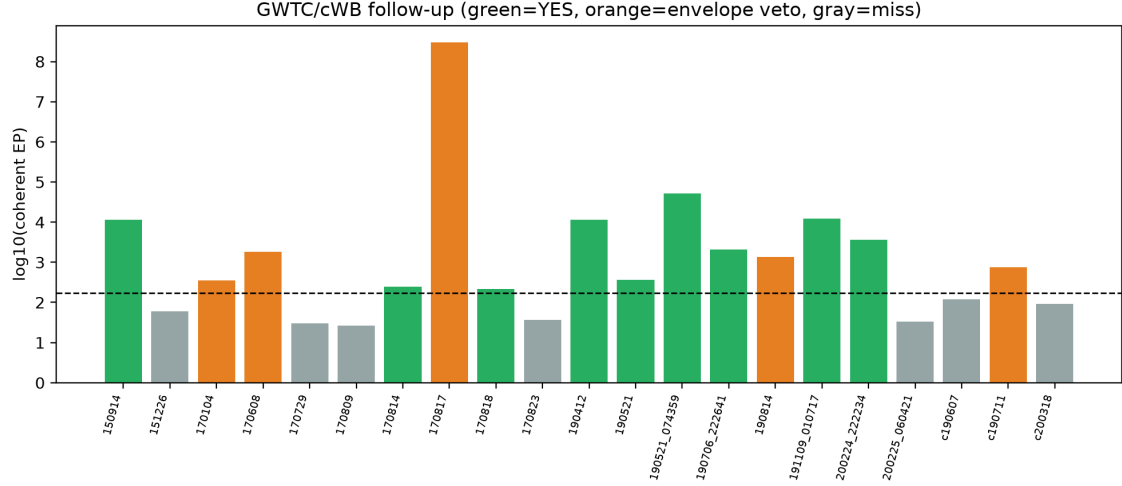


Figure 3: Follow-up coherent EP by event. Green: production YES; orange: envelope veto; gray: miss.

7 Reproducibility

Frozen artifacts are stored under `docs/freeze/current/` with checkpoint SHA256 prefix `55ce7637...` (label `2026-07-paper-v1`). Reproduce with:

```
cd ml-engine
python -m app.evaluation.aresgw_class --train
python -m app.prove --freeze --freeze-label 2026-07-paper-v1
```

8 Conclusion

Generative noise subtraction followed by template-free residual search can recover unknown-morphology bursts at controlled false-alarm rate in a reproducible research prototype. Keno is released as a software contribution for unmodeled residual search — complementary to existing BBH classifiers — for audit and extension, not as a finished gravitational-wave pipeline.

Data availability

Code and freeze bundle accompany this manuscript. Large evaluation CSVs are hashed in the freeze manifest.

References

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- A. E. Koloniari et al., New gravitational wave discoveries enabled by machine learning, *Mach. Learn.: Sci. Technol.* (2025).
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